

Response to the Referee

We thank the referee for a careful and constructive report. The central message — that the manuscript read as a machine-learning benchmarking exercise and needed astrophysical answers — was well taken, and the revision is substantially strengthened as a result. We have (i) replaced the "finite intermediate-mass window" framing with a physically supported **upper-bounded gas-channel regime**, (ii) added a sequence of mechanism analyses that identify *what kind* of gas information the signal carries and *why* it terminates at high mass, (iii) scoped the cross-simulation comparison to a matched control, and (iv) stated honestly the two points that are genuine data limits rather than oversights. All new analyses reuse the paper pipeline (5-fold ridge CV R^2 for growth, logistic CV AUC for quenching, paired bootstrap 95% CIs).

Summary of principal changes

- **Reframing / title.** The lower edge at $\log M_\star \approx 9.55$ is *not* a sharp physical

boundary (Comment 4); the defensible result is an upper-bounded regime in which internal gas state outpredicts assembly history up to $\log M_\star \approx 10.55$. The title and abstract are revised accordingly; "finite intermediate-mass window" is retired.

- **New physical mechanism (addresses the general critique + Comments 1, 6).** The

residual gas signal is **fuel-amount information**: it survives controls for the current star-formation state *and* stellar mass, it is carried by gas amount rather than depletion-time (efficiency), the reservoir is largely (but not fully) shaped by halo assembly history, and near the upper transition black-hole/quenching state becomes the relevant predictor.

- **Honest data limits.** Direct AGN feedback energy and a higher-resolution

convergence run are not available in our products; both are now stated as such rather than implied.

Detailed responses

Comment 1 — physical context for the window boundaries; AGN feedback

We added population profiles (gas fraction, sSFR, quenched fraction, BH-state proxies) as a function of stellar mass with the window edges marked, for both TNG and SIMBA.

Upper boundary (~ 10.55) is physically supported. Across the upper edge the median log sSFR drops from -9.63 (inside) to -10.96 , the quenched fraction rises from 0.035 to 0.498, and median BH mass rises by $+1.51$ dex — i.e. the edge coincides with the onset of AGN-associated quenching. These population profiles (gas fraction, sSFR, quenched fraction, and BH mass versus

stellar mass, with the upper transition marked) are now shown as an appendix figure (population diagnostics, not predictive-model inputs).

Lower boundary (~ 9.55) is not a physical threshold (see Comment 4): it is a floor-encoding / resolution-limited measurement edge. We therefore reframe the result as an upper-bounded regime.

AGN feedback energy is genuinely unavailable. An exhaustive scan of all local CAMELS catalog products confirms they are SUBFIND group catalogs and merger trees only, with no particle-level black-hole data and no energy-injection field. We therefore use BH mass, BH accretion rate (BHMDot) and cumulative wind mass (WindMass) only as black-hole / feedback **state** proxies, and we do not claim a direct feedback-energy measurement; recovering the injected energy would require the CAMELS snapshot black-hole-particle data and is noted as future work. To go beyond proxies we instead added the mechanism tests below (depletion time, BH absorber), which address **why** the gas signal terminates at high mass using the available state information.

Comment 2 — the unresolved $\sim 71\%$ of the absorbed internal signal

We expanded the absorption analysis well beyond the original three ablations: every L3 feature is removed individually, in physically motivated groups, and in all group pairs, with a greedy cumulative path, a non-linear (random-forest / gradient-boosting) L3 comparison, and a redundancy/PCA characterisation. The reproduction matches the published L1→L3 absorption exactly (weak-L1 internal marginal 0.308; full-L3 internal marginal 0.135; absorbed 0.173). The strongest single separated group is the half-mass epoch (+0.022, $\approx 16\%$ of the full-L3 internal marginal); individually ablating the early-accretion features ($\Delta \log M$ over 1–2 Gyr, the sl4/sl8 lookbacks) recovers $\leq 7\%$ each, and because the assembly features are collinear these recoveries overlap rather than add; the 13 L3 features require 8 principal components for 90% of their variance; and flexible non-linear baselines absorb no more than ridge. We therefore report the verdict as a **distributed, correlated assembly-history manifold** rather than a single missing absorber. We state explicitly in the manuscript (§3.7):

We interpret the unresolved component as distributed, correlated assembly-history structure rather than a single missing absorber; because a richer feature set could absorb additional signal, we treat this as a decomposition limit of the present L3 control rather than a claim of fundamental irreducibility.

Comment 3 — asymmetric TNG (L3) vs SIMBA (L1) comparison

We agree the original TNG-L3-vs-SIMBA-L1 comparison was asymmetric. We now compare the two simulations **only at the shared L1 (static-halo) control**, separately in the growth (R^2) and quenching (AUC) channels:

L1 internal-family marginal	TNG	SIMBA
Growth (R^2)	+0.245 [+0.205, +0.293]	+0.080 [−0.144, +0.175] (n.s.)
Quenching (AUC)	+0.060 [+0.032, +0.081]	+0.141 [+0.096, +0.188]

At a matched L1 control, SIMBA carries its beyond-gravity internal signal in the quenching channel and TNG in the growth channel — a genuine, but strictly L1-scoped, cross-model contrast (see Fig.

7). We have removed the TNG-L3-vs-SIMBA-L1 ($\sim 3\times$) comparison, report only the appropriate L1-vs-L1 quenching ratio ($\sim 2.4\times$), make no claim of a SIMBA L3-controlled residual (SubLink-equivalent trees are unavailable at CAMELS-SIMBA resolution), and retain the TNG growth/L3 result as the paper’s primary finding.

Comment 4 — box size / volume, environment, and the lower edge

Lower edge. Below $\log M_\star \approx 9.55$ the raw gas-only marginal is destabilised by a small number of unresolved objects whose gas feature is pinned at the catalog floor (the full-sample estimate is $+0.005$ with a meaningless interval, $[-0.279, +0.010]$). Removing that tail gives $+0.061$, and — crucially — **winsorizing the floor-encoded gas feature without removing any object recovers $+0.057$** , matching both the cut and the onset band (see Appendix Fig. 13). The lower edge is therefore a floor-encoding / resolution-limited measurement edge, not a sharp physical threshold; we disclose that the affected objects are also low-growth (mean $\Delta \log M_\star -0.11$ vs $+0.40$). The robust result is an upper-bounded regime with an explicit low-mass resolution caveat; a dedicated higher-resolution convergence comparison is unavailable to us and is noted as such.

Environment / volume. We agree the environmental null is volume-specific: in a $25 \text{ h}^{-1} \text{ Mpc}$ box large-scale modes are coarsely sampled. We now state the environmental result as specific to this volume and feature set rather than as a general finding.

Comment 5 — linearity of the ridge model

We re-ran the mid-mass internal-gas test with non-linear models, fitting the 13-feature assembly-history baseline *itself* with the same random-forest / gradient-boosting model so the baseline is free to capture non-linear assembly-history structure (not only the internal model). The internal-family marginal survives: ridge $+0.139$ [$+0.119, +0.153$], random forest $+0.120$ [$+0.102, +0.132$], gradient boosting $+0.119$ [$+0.100, +0.135$]; gas mass remains the top internal feature in all three. A more flexible, non-linear assembly-history baseline therefore does not absorb the gas signal. The result is not an artifact of linearity; these values (ridge $+0.139$, random forest $+0.120$, histogram gradient boosting $+0.119$; gas mass top in all three) are now reported explicitly in the manuscript robustness subsection (§3.9).

Comment 6 — $f_\star$ (integrated) vs gas (instantaneous), and §3.7.3

Adding $f_\star = M_\star/M_{\text{sub}}$ to the L3 control absorbs most of the halo family’s surviving marginal ($0.083 \rightarrow 0.017$) while a positive internal marginal remains ($0.135 \rightarrow 0.086$), consistent with the halo residual tracking accumulated star-formation efficiency and the internal residual tracking an instantaneous fuel state. We now make the integrated-vs-instantaneous distinction explicit and, more importantly, identify the instantaneous quantity: it is the **amount** of gas, not the rate of star formation or the conversion efficiency. Controlling for the current SFR/sSFR *and* stellar mass, gas mass still adds $+0.022$ [$+0.015, +0.029$] (intermediate) and $+0.069$ [$+0.053, +0.086$] (low); adding the depletion time $t_{\text{dep}} = M_{\text{gas}}/\text{SFR}$ (stellar-mass-free) contributes nothing once gas amount is included, while gas amount adds beyond it. A coarsened-exact matched-pair test corroborates a positive gas→growth effect at low mass ($+0.14$ dex at fixed mass + halo + assembly history). These mechanism results are shown in Fig. 6 and Appendix Figs. 8–12. We note that

the loaded CAMELS group-catalog products provide total subhalo gas mass only — no phase-separated cold or star-forming gas-mass field is present — so we frame the prediction in terms of total gas-reservoir content and flag cold/ star-forming phase separation as requiring particle-level or value-added data (now stated as a caveat in §4.6). The xGASS/xCOLD GASS test should accordingly be read as probing whether the observable cold-gas reservoir traces the same residual fuel channel, not as a one-to-one measurement of the total-gas simulation predictor.

Comment 7 — the halo L3 marginal triangles

These are drawn as discrete triangles at the three individually-significant windows (9.85, 10.15, 10.35 dex) precisely because the L3 halo signal is non-contiguous. We add to the Fig. 2 caption:

The halo L3 marginal is shown as discrete triangles, with no connecting line, because the L3-controlled halo signal is non-contiguous (only the three marked windows are individually significant); a connecting line would imply a continuous trend the data do not support.

Comment 8 — near-unpredictability of halo growth

We confirm and now interpret this result (see the revised halo-growth paragraph, §3.1). Predicting halo growth ($\Delta \log M_{\text{sub}}$) from any family gives R^2 of order 0.01–0.03 (each family $\lesssim 0.016$ in the intermediate window) — an order of magnitude below stellar growth (internal-family $R^2 \sim 0.2$ –0.3). We add:

That halo growth is near-unpredictable from any $z \approx 0.77$ family is itself informative: the dark-matter accretion a subhalo will undergo by $z = 0$ is essentially not encoded in its individual structural, assembly-history or environmental properties at the earlier epoch. Consequently no early property serves as a proxy for future halo growth, and a galaxy’s baryonic future (stellar growth) is substantially more predictable from its early internal state than its gravitational future. (We scope the $R^2 \leq 0.015$ statement to the intermediate window; in the whole sample the halo family reaches ≈ 0.03 , still negligible.)

We believe these changes address the referee’s concerns directly and convert the manuscript from a benchmarking exercise into a physically interpreted result. We thank the referee again for comments that materially improved the paper.